

Better Executives Work at Better Firms: Identifying Talent Sorting from Mobility Networks

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ABSTRACT

Executive talent and firm productivity are complementary, so the allocation of executives across firms can affect aggregate output. We infer talent and productivity from business outcomes and executives' moves across firms. Outcomes at firms connected by short sequences of executive moves covary more strongly than outcomes at firms connected by longer sequences. How quickly this covariance declines with distance identifies the correlation between executive talent and firm productivity. We apply the method to administrative records covering close to the universe of Hungarian corporations and their executives from 1990 to 2022. The data include more than one million firms and 1.3 million executives. We estimate a correlation of 0.5 between executive talent and firm productivity. In an accounting calculation, we remove this covariance while keeping the separate distributions of firm productivity and executive talent and the match shocks unchanged. Average revenue falls by 13%.

1 INTRODUCTION

The allocation of talent across firms can affect aggregate output [19, 25]. Suppose that firm i has productivity A_i and executive m has talent Z_m . Their match contributes $A_i Z_m$ to firm output. Aggregate output sums these products across executive–firm matches,

$$Y = \sum_{(i,m) \in \mathcal{M}} A_i Z_m.$$

Here $\mathcal{M}$ denotes the set of executive–firm matches. Holding the distributions of A_i and Z_m and the number of matches fixed, aggregate output rises when talented executives work at productive firms, or equivalently when talent and productivity are more positively correlated. We call this firm–executive sorting. How strong is this correlation?

Measuring this correlation across the whole economy requires records beyond publicly listed firms. Studies use executives' moves across firms and business outcomes to infer their persistent contributions [6, 14, 22]. Other studies measure how much firm performance varies with the identity of the chief executive officer (CEO) [26]. Short tenures make persistent contributions hard to

distinguish from chance [16]. We measure sorting across the broad corporate sector.¹

Methods for linked worker–firm data use workers' moves between firms to separate persistent worker and firm effects [1]. Executives, however, typically work at only a few firms, and firms change executives infrequently. The resulting estimates of individual firm and executive effects are imprecise.

Leave-out estimators remove each moving worker in turn and require the other workers and firms to remain connected through other moves [23]. Among executives, only firms connected by several routes of executive moves satisfy this requirement. Those firms may differ systematically from firms with stable leadership, making the selected sample unsuitable for an economy-wide analysis. We instead estimate the variances of firm and executive effects and the covariance between them. The method uses every retained group of firms and executives linked by observed appointments, including small groups.

The basic comparison begins with a chain of two executives and two firms,

$$m_1 - i_1 - m_2 - i_2.$$

First, executive m_2 replaces executive m_1 at firm i_1 , so the change in business performance contains information about their relative talent. Executive m_2 then moves to firm i_2 , where performance provides another observation related to m_2 's talent. This path connects the outcome for m_1 at i_1 with the outcome for m_2 at i_2 , although the outcomes involve different executives and firms. Under positive sorting, executives who lead the same firm tend to have similar talent, while firms led by the same executive tend to have similar productivity. Business outcomes at the two ends of the chain should therefore covary positively. This covariance should become weaker along longer chains. We call that decline covariance decay. Its rate identifies the magnitude of the correlation between executive talent and firm productivity. Comparisons with a firm at one end and an executive at the other help determine whether the correlation is positive or negative. The likelihood combines this evidence with estimates of how much firm and executive quality vary.

¹Executive pay provides a secondary motivation. Competitive assignment models explain compensation differences by pairing more talented executives with more productive firms [17, 31].

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To interpret this covariance as evidence about persistent firm and executive qualities, we assume that transitory business shocks are independent across firms. The estimators allow shocks within the same firm to be correlated.

We represent executive appointments as a network linking firms and executives. Each firm’s unobserved productivity depends directly only on the talents of its executives, and each executive’s unobserved talent depends directly only on the productivities of the firms they lead. Once these neighboring qualities are known, more distant qualities provide no additional information. This structure is a Gaussian Markov random field (GMRF) [28]. The parameter ρ governs dependence along network paths. On a network without cycles, dependence declines geometrically with path length, just as correlation in a first-order autoregressive (AR(1)) time series declines with time lag.

This network model echoes work on family networks and classical quantitative genetics [12, 15].

A simple account of how firms and executives match generates the same model. Before matches form, firm productivity and executive talent are independent in the population. Complementarity makes more productive firms and more talented executives more likely to match. Among observed matches, productivity and talent are therefore correlated, and business outcomes also contain transitory shocks. Because observed links are sparse, the derivation retains the terms for firm–executive pairs that match and drops the terms for eligible pairs that do not. This approximation produces the GMRF distribution on the observed appointment network. Materials and Methods derives the approximation. Under additional assumptions, the supplement compares it with a complete logistic model that assigns a link probability to every possible match within a fixed, finite set of eligible pairs. Both models treat links independently within that set. A market-clearing assignment would require a different model. The estimate therefore describes sorting among the observed links, conditional on which links formed.

We apply the method to Hungarian administrative records of firms and registered executives from 1990 to 2022. We allow shocks from the same firm to be correlated in three ways: we model their serial correlation, average outcomes within firms, or use only covariance comparisons between different firms. These approaches yield estimates of ρ between 0.518 and 0.555, compared with 0.379 when match shocks are independent. To measure how sorting contributes to revenue, the fitted model gives a covariance between productivity and talent for each firm–executive link with a revenue observation. We average twice these covariances across links, giving each link equal weight. We then calculate average revenue after setting this covariance to zero while holding the firm and executive distributions and match shocks fixed.

Section 2 explains how the registry records become the executive-mobility network and the estimation sample. The following sections present the sorting estimates, calculate their contribution to revenue, and state the assumptions required for interpretation. Materials and Methods and the supplement derive the matching model, explain the computation, and compare the estimators.

2 DATA AND THE EXECUTIVE MOBILITY NETWORK

Hungarian administrative records let us study executive sorting beyond publicly listed firms. We combine a registry of executives authorized to represent a company, maintained by corporate courts, with annual accounting records for all firms that use double-entry bookkeeping [11, 20, 21]. The panel spans 1990 to 2022. Materials and Methods explains how we link records that refer to the same firm or executive and select the estimation sample.

The registry covers every officer authorized to represent a company. We identify chief executives from titles where available. Where titles are unavailable, we use continuity and the number of registered officers. This procedure defines the executive population used in the network.

We represent the registry as a bipartite graph: firms and executives are the two types of nodes, and we draw one graph edge for each unique firm–executive pair. Repeated job spells for the same pair count once. A path alternates between firm and executive nodes. A cycle is a route that returns to its starting node without repeating an edge. A connected component contains every node joined by paths. Table 1 reports these connections before we remove any edges. The graph contains 1,036,231 firms and 1,304,580 executives joined by 1,937,552 graph edges. For estimation, we later combine revenue records with these graph links.

The graph is sparse and fragmented, with 514,085 connected components. Its largest component, the giant component, contains 386,015 executives and 341,650 firms joined by 823,158 graph edges. A bridge is an edge whose removal splits a component; 81.8% of edges in the full registry graph are bridges. These numbers describe the graph before we remove cycle-closing edges. The estimator section reports how many grouped revenue observations enter the analysis.

The mean shortest path between sampled executive pairs in the giant component is 19.5 graph edges. Even this large component therefore supplies relatively few short paths for comparing executives.

Figure 1 illustrates the graph, including a path, a cycle, and several small components.

2.1 Coverage beyond the largest component

Our estimation sample includes every connected component that survives the sample restrictions. The likelihood contains one term for each component, so every retained group enters separately. An isolated firm–executive pair helps measure the total variation in revenue. Paths that connect several firms and executives show how covariance changes with distance and therefore help measure sorting.

We begin with the registry graph, apply the sample restrictions, and remove some edges that close cycles. Removing these edges ensures that the model defines a valid covariance matrix at the estimated correlation. We call this procedure spectral pruning. The pattern of graph connections alone determines which edges are removed.

We measure the influence of smaller components by adding roughly 426,000 of them to a graph containing the larger components. This expansion nearly doubles the node count and raises the

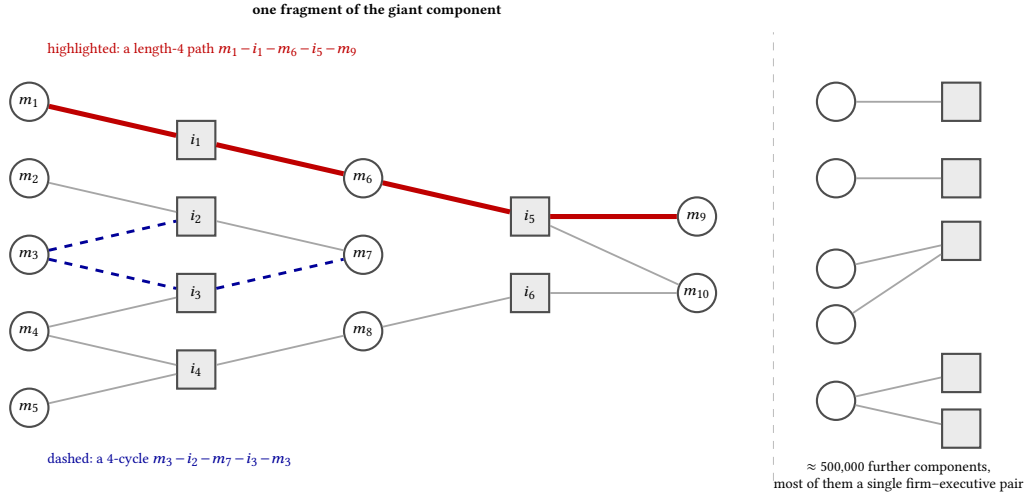


Figure 1: The bipartite executive-firm mobility graph. Circles are executives, squares are firms, and edges are unique firm-executive links. Left: a schematic fragment of the giant component, with one length-4 path highlighted in red and one 4-cycle dashed in blue. Right: examples of small components. The drawing is schematic; Table 1 reports the empirical counts for the unpruned registry graph.

Table 1: Descriptive statistics of the unpruned firm-executive network after sample construction. Edges are unique firm-executive pairs, not spells or grouped outcome observations. Bridges and their share refer to this binary graph. Path length counts edges between sampled executive pairs in the giant component.

	Value
<i>Full network</i>	
Executives	1,304,580
Firms	1,036,231
Firm-executive edges	1,937,552
Connected components	514,085
Bridges	1,584,810
Bridge share of edges	81.8%
<i>Giant component</i>	
Executives	386,015
Firms	341,650
Edges	823,158
Share of executives	29.6%
Share of firms	33.0%
<i>Giant component paths</i>	
Mean shortest path, executive pairs	19.5

number of edges by 59%. Across two estimators and several choices of cycle-closing edges, adding the smaller components changes $\hat{\rho}$ by at most 0.011. Before estimating this comparison, we subtract the mean outcome for each combination of year and industry and adjust for the number of graph links of the firm and executive.

Table 2 compares firms inside and outside the giant component of the unpruned graph. The giant contains 33.0% of firms and 29.6% of executives, but 55.9% of employment. Mean employment is 9.1

Table 2: Composition of the giant component versus the remaining components of the unpruned sample network. Employment is a firm's average headcount over observed years; revenue is in logs. Manufacturing is TEAOR08 section C. Match rows count executive entries before grouping simultaneous executives into one outcome. Founder-classified shares use the recorded category 1 over those rows, not a singleton-graph proxy. These counts are not likelihood observation counts.

	Giant	Other	All
<i>Counts</i>			
Firms	341,650	694,581	1,036,231
Executives	386,015	918,565	1,304,580
Match rows	1,116,741	1,409,277	2,526,018
<i>Shares of total (%)</i>			
Firms	33.0	67.0	100.0
Executives	29.6	70.4	100.0
Employment	55.9	44.1	100.0
<i>Firm characteristics</i>			
Employment, mean	9.1	3.5	5.4
Employment, median	1.4	1.2	1.2
Log revenue, mean	9.37	8.65	8.89
Manufacturing	8.5%	9.5%	9.2%
Founder-classified rows	46.8%	72.1%	60.9%

inside and 3.5 outside; the corresponding medians are 1.4 and 1.2. Manufacturing accounts for 8.5% and 9.5% of firms, respectively. Restricting the sample to the giant component would therefore over-represent larger firms. These aggregates pool detailed industries and locations.

The registry classifies some records as founders. Founder records are more common outside the giant component: they account for 72.1% of records outside and 46.8% inside. This difference describes how founder records are distributed across connected components.

Firm size rises across the smallest components and then flattens. Mean employment increases from 2.76 in components containing one firm–executive pair to 5.87 in components with 11–50 nodes, then remains at 5.84 in larger non-giant components. Mean log revenue follows the same pattern, rising from 8.33 to 9.22 and then remaining at 9.23.

Small components include firms with little observed executive mobility. Component size measures network connectivity; founder status, industry, and geography remain separate characteristics.

3 MEASURING TALENT FROM MOBILITY

Let y_{im} denote log real revenue at firm i when executive m serves it. In the simplest case with one executive, the performance model is

$$y_{im} = a_i + z_m + \varepsilon_{im}.$$

The firm effect a_i captures persistent revenue differences across firms, and the executive effect z_m captures persistent revenue differences across executives. The match shock ε_{im} captures transitory features of their time together. We interpret the persistent effects as firm productivity and executive talent under the assumption that persistent covariance along executive-mobility paths comes from these two qualities.

We divide each firm’s history into match clusters: intervals during which its set of executives is constant. Each cluster contributes one outcome containing the firm effect once, the average of the executive effects, and one match shock.

The three-edge chain in the Introduction provides the simplest example of how the model measures sorting. The simplest case is a graph with no cycles, called a forest. On such a graph, the covariance between the outcomes at the ends of this chain is

$$\text{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) = \rho(\sigma_a + \rho\sigma_z)(\sigma_z + \rho\sigma_a).$$

The formula describes both the level of covariance and how quickly covariance declines with distance. The parameters σ_a and σ_z describe how much persistent firm and executive quality vary, while ρ also determines the rate of decline. Consider two paths whose endpoints have the same arrangement of node types: firm–firm, executive–executive, or firm–executive. Adding two graph edges to a path multiplies covariance by ρ^2 . Comparing such paths separates the rate of decline from the initial level of covariance.

The empirical graph contains cycles, so two nodes can be connected by several routes. These additional routes change the covariance between outcomes. The likelihood therefore calculates covariance from the actual pruned graph, while the forest formula provides intuition for its level and decay.

Successive outcomes at the same firm can also covary through correlated transitory shocks. We use three approaches to separate these shocks from persistent firm and executive qualities. The first approach chooses parameters so that model covariances match observed covariances between outcomes at different firms; we call it minimum distance. Because it compares outcomes from different firms, shocks within a firm may have any covariance. The second

approach averages all match-cluster outcomes within each firm and fits the model to those firm averages; shocks within a firm may again have any covariance. The third approach orders each firm’s match clusters over time and assumes that correlation between their shocks declines geometrically with the distance between them in that sequence. This AR(1) likelihood estimates one common rate of decline for all firms jointly with sorting.

4 SORTING BETWEEN EXECUTIVES AND FIRMS

Outcomes under successive executives at the same firm contain two sources of covariance: persistent firm quality and correlated transitory shocks. A likelihood that assumes independent transitory shocks can attribute some covariance from those shocks to persistent firm and executive differences. Under that assumption, the estimated sorting parameter is $\hat{\rho} = 0.379$. We use it as the baseline for three methods that allow shocks from the same firm to be correlated.

Our preferred likelihood orders each firm’s match clusters over time and assumes that their transitory shocks follow one common AR(1) process. The parameter η is the correlation between shocks in adjacent match clusters, and correlation declines geometrically as clusters become farther apart in the sequence. The same process and value of η apply to every firm. The model uses every match cluster with a revenue outcome and jointly estimates $\hat{\rho} = 0.518$, $\hat{\sigma}_a = 1.078$, $\hat{\sigma}_z = 0.240$, $\hat{\sigma}_\varepsilon = 1.721$, and $\hat{\eta} = 0.506$. The data contain $K = 1,666,857$ outcome observations from $N_f = 1,036,228$ firms and $N_m = 1,304,563$ executives. The graph also retains 191,659 links that connect firms and executives but have no revenue observation. For this graph, the model defines a valid covariance matrix when $|\rho| < 0.679$. We restrict estimation to $|\rho| \leq 0.666$ for numerical safety; the estimate of 0.518 lies inside both bounds.

Table 3 compares the independent-shock baseline, minimum distance, firm averaging, and the AR(1) model on the same graph.

On a forest, ρ is the correlation for every linked firm–executive pair. On the fitted graph, cycles make variances and covariances depend on the connections surrounding each link. For links with a revenue observation, Table 3 averages $\text{Var}(a_i)$, $\text{Var}(z_m)$, and $2\text{Cov}(a_i, z_m)$, giving each link equal weight. It then reports $V_{\text{cross}} / (2\sqrt{V_{\text{firm}} V_{\text{executive}}})$ as the correlation for linked pairs. The parameter ρ remains the precision parameter that governs path dependence.

Minimum distance estimates the parameters by bringing model covariances close to observed covariances along selected paths between different firms. It estimates $\hat{\rho} = 0.555$, which is also the correlation between firm and executive effects for linked pairs under this estimator. Because its comparisons use outcomes from different firms, shocks within a firm may have any covariance. Its reported total of 2.165 sums the firm variance, executive variance, and twice their covariance. The likelihood totals also include the variance of match shocks.

The firm-average likelihood first averages the match-cluster outcomes within each firm and then fits the model to one average per firm. It permits any covariance among shocks from the same firm. The estimated sorting parameter is $\hat{\rho} = 0.537$, and the correlation between firm and executive effects for linked pairs is 0.544.

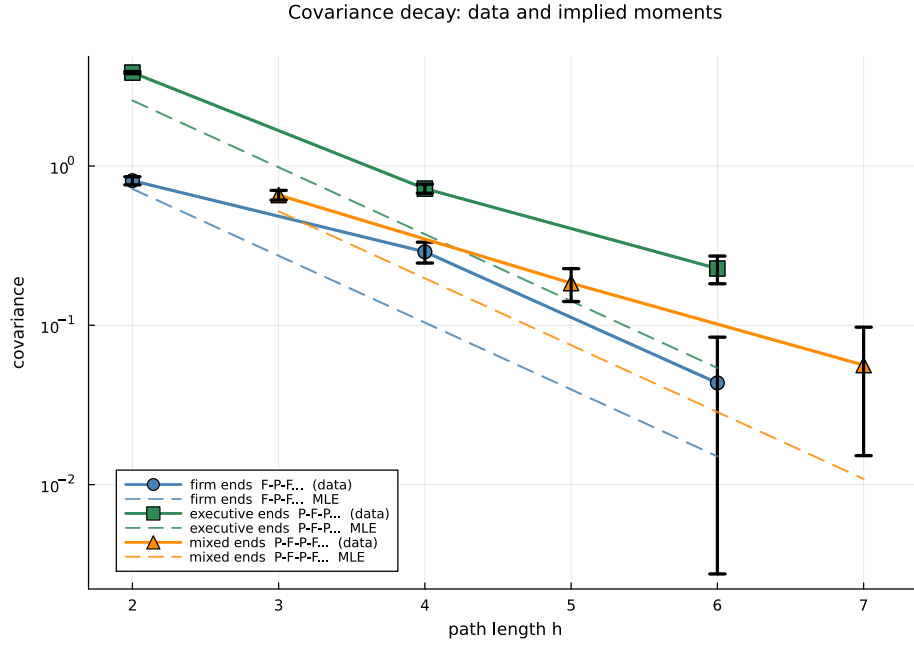


Figure 2: Covariance along network paths. For each path length and arrangement of endpoint types, we center revenue outcomes at the left and right endpoints separately. Each marker is the equal-weighted average of the products of these centered outcomes across retained outcome pairs; bars are 95% confidence intervals. Colors and marker shapes distinguish paths with two firm endpoints, two executive endpoints, or one endpoint of each type. Each retained path is the shortest route between its endpoints and is sufficiently separated from alternative routes. The same underlying sequences of executive moves supply each distance and endpoint comparison. We fit the model that assumes independent transitory shocks on the full pruned graph. Dashed curves show the covariances that its forest formulas predict for the sampled paths. Supplementary Material 1 describes path selection and an additional estimate based on the plotted covariances. Table 3 reports the main minimum-distance estimate, which uses a different set of paths and covariances between outcomes at different firms. The vertical axis uses a logarithmic scale.

Table 3: Firm-executive sorting under alternative assumptions about shocks from the same firm. All columns use the same graph, with one edge per linked firm-executive pair and the same cycle-closing edges removed. We subtract each calendar year’s mean from log real revenue. The likelihood columns group intervals with the same set of executives into match clusters; the firm-average column then averages the resulting outcomes within each firm. Minimum distance matches observed and model covariances along selected paths between different firms. The independent-shocks column assumes that shocks in different match clusters are independent. For every firm-executive link with a revenue observation, the fitted model gives a firm variance, an executive variance, and twice their covariance. The reported variance rows average these quantities across links, giving each link equal weight. Because the graph contains cycles, the connections surrounding each link also affect these variances and covariances.

Quantity	Independent shocks	Minimum distance	Firm averages	AR(1) shocks
ρ	0.379	0.555	0.537	0.518
V_{firm}	2.222	1.974	1.042	1.178
$V_{\text{executive}}$	0.090	0.013	0.085	0.058
V_{cross}	0.341	0.178	0.324	0.275
V_{ε}	1.595	—	2.228	2.634
V_{total}	4.249	2.165	3.679	4.145
Correlation for linked pairs (a, z)	0.381	0.555	0.544	0.525

The three methods that allow shocks from the same firm to be correlated estimate correlations of 0.52–0.55 between the firm and executive effects of linked pairs. Because cycles make the variances

and covariances entering this ratio depend on each link’s surrounding connections, the reported correlation can differ slightly from ρ .

These estimates vary less across methods than the estimated firm, executive, and shock variances.

4.1 Economic significance

At the preferred AR(1) estimates, persistent differences across firms account for 28.4% of total log-revenue variance, and persistent differences across executives account for 1.4%. Covariance between the firm and executive effects accounts for 6.6%, while transitory shocks account for 63.5%. The firm effect is the largest persistent source of variance, while transitory shocks account for the largest share overall. The three methods agree more closely about sorting than about this division of variance. The covariance term measures how much the firm and executive effects move together; establishing whether allocation causes firms to perform better requires separate evidence.

We translate this covariance into average revenue with an accounting calculation. Firm productivity and executive talent are complementary and persistently heterogeneous [30]. Let n denote the number of executive–firm matches. Under the model’s log-normal distribution, with $A = \exp(a)$ and $Z = \exp(z)$,

$$\begin{aligned} Y &= n \mathbb{E}[AZ \exp(\epsilon)], \\ \frac{Y}{n} &= \exp\left\{\mu_a + \mu_z + \mu_\epsilon + \frac{1}{2}(V_a + V_z + V_\epsilon + V_{\text{cross}})\right\}, \\ V_{\text{cross}} &= 2 \text{Cov}(a, z). \end{aligned}$$

For each firm–executive link with a revenue observation, the fitted model gives a covariance between the firm’s productivity and the executive’s talent. The term V_{cross} is twice this covariance, averaged across links with equal weight. To measure its contribution to average revenue, we set this term to zero while keeping the separate distributions of firm productivity and executive talent and the match shocks unchanged. The log-normal formula then changes by $\exp(-V_{\text{cross}}/2) - 1$. At the preferred estimates, $V_{\text{cross}} = 0.275$, so average revenue falls by about 0.138 log points, or 12.9%. This calculation changes only the covariance. Setting $\rho = 0$ in the model would also change the separate firm and executive variances because the graph contains cycles.

5 DISCUSSION

All three methods that allow shocks from the same firm to be correlated find a positive correlation between the persistent revenue components attributed to firms and executives in linked pairs. Their agreement on sorting does not imply agreement on every variance component. It also does not establish that a change in executive identity causes the estimated revenue difference.

In the preferred decomposition, sorting covariance accounts for 6.6% of total log-revenue variance. Removing that component while holding the estimated marginal distributions and match shocks fixed lowers average revenue by 12.9% under the log-normal accounting benchmark.

Specification and scope. The model assumes that the unobserved qualities of unlinked firms and executives are independent once we condition on the qualities of their directly connected neighbors. Industry or geographic similarity can also generate covariance along mobility paths. Under this restriction, the path-covariance estimates identify sorting. Because the neighboring qualities are unobserved,

the estimates do not separately identify covariance generated by firms and executives matching within the same industries or places.

Executive changes can respond to firm shocks, and selection into executive roles can correlate with unobserved firm characteristics. Owner-managed firms can combine executive skill with ownership decisions, while an executive may leave a persistent effect after departure. Our design identifies sorting in persistent revenue components rather than identifying these channels separately. We therefore interpret the accounting exercise within the observed executive–firm sector and the maintained model, not as an economy-wide welfare calculation.

Leave-out fixed-effects methods provide a complementary benchmark [23]. The supplement first compares estimators in a dense worker–firm sample, where mobility supports leave-out estimation, and then compares the estimators on the same sparse executive sample. The executive comparison assumes independent match shocks, so it isolates differences between estimators under that shock model rather than differences caused by serial dependence within firms [32].

Research on executive and manager effects documents heterogeneity in firms and individuals [4–7, 14, 24, 26]. Our contribution is to measure their allocation on a sparse mobility network while allowing for correlated match shocks. Event studies of executive changes address a different source of variation [24]. Equilibrium matching models derive sorting from complementarities, preferences, and search frictions [3, 13, 18, 29]; our conditional network model complements that work. The same measurement approach may also apply to buyer–supplier and inventor networks.

6 MATERIALS AND METHODS

6.1 Population types, performance, and link formation

Before links form, we model firm types a_i and executive types z_m as independent draws from zero-mean Gaussian populations with variances ω_a^2 and ω_z^2 . These types are not observed directly, so we refer to them as latent types. We stack them in $\mathbf{x} = (\mathbf{a}^\top, \mathbf{z}^\top)^\top$. Their population density $p_0(\mathbf{x})$ has diagonal inverse covariance matrix $\mathbf{Q}_0$. Independence is a benchmark for the selection argument, not an empirical finding about observed matches.

Persistent firm quality and executive skill are complementary inputs to performance [25]. With $A_i = \exp(a_i)$ and $Z_m = \exp(z_m)$, revenue on a single-executive match is

$$Y_{im} = A_i Z_m \exp(\epsilon_{im}).$$

With fixed supplies and no other assignment constraints, complementarity favors pairing higher-type firms with higher-type executives, an allocation called positive assortative matching. Observed sorting can differ because firms and executives have other matching options and because frictions make some links less likely to form; the production equation alone does not determine the assignment.

Taking logs decomposes the firm outcome into persistent firm and executive effects and a match shock:

$$y_{im} = a_i + z_m + \epsilon_{im}. \quad (1)$$

We assume that match shocks are independent of the latent types and independent across firms; they may be serially correlated within

a firm. We specify the shock process after deriving the distribution of selected types.

For a potential firm-executive pair, let $\kappa + h_{im}(\mathbf{x})$ be the systematic index that determines its link probability, where

$$h_{im}(\mathbf{x}) = \frac{1}{\tau} \left[a_i z_m - \alpha_1 a_i - \zeta_1 z_m - \frac{\alpha_2}{2} a_i^2 - \frac{\zeta_2}{2} z_m^2 \right]. \quad (2)$$

The parameter $\tau > 0$ scales link-formation frictions. The cross term expresses complementarity; the linear and quadratic terms allow outside options to vary with type. Conditional on types, links form independently with logistic probabilities $p_{im} = \exp(\kappa + h_{im}) / [1 + \exp(\kappa + h_{im})]$. This is a data-generating process for independent link selection, not a one-to-one matching equilibrium.

To state what the approximation omits, consider a fixed, finite set D of eligible firm-executive pairs that contains the observed unique links E . Conditioning on the complete link pattern gives three factors: the population density, the probabilities of formed links, and the probabilities of absent links,

$$\begin{aligned} p^F(\mathbf{x} \mid E, D) &\propto p_0(\mathbf{x}) \prod_{d \in E} p_d(\mathbf{x}) \prod_{d \in D \setminus E} [1 - p_d(\mathbf{x})] \\ &= p_0(\mathbf{x}) \frac{\exp\{\sum_{d \in E} [\kappa + h_d(\mathbf{x})]\}}{\prod_{d \in D} [1 + \exp(\kappa + h_d(\mathbf{x}))]}. \end{aligned} \quad (3)$$

When the probability of a link is small, we approximate $p_d(\mathbf{x})$ by $\exp(\kappa + h_d(\mathbf{x}))$, whose log contribution is quadratic. Retaining only these positive-link contributions also drops the denominator shared by all pairs in equation (3). The resulting normalized density is

$$p^+(\mathbf{x} \mid E) \propto p_0(\mathbf{x}) \exp\left\{\sum_{d \in E} h_d(\mathbf{x})\right\} \propto \exp\left(-\frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{b}^\top \mathbf{x}\right), \quad (4)$$

provided $\mathbf{Q}$ is positive definite, so that $\mathbf{Q}^{-1}$ is a valid covariance matrix. The intercept κ cancels from this conditional positive-link density; the fitted model does not estimate the absolute link rate.

An eligible pair that does not link—for example, an executive not serving as Apple’s CEO—may convey little information about that executive. Many such pairs can nevertheless have a non-negligible aggregate contribution. Supplementary Material 6 bounds the omitted denominator when the graph condition remains valid after adding any one eligible edge and the relevant moments have finite bounds. Bounds on likelihood derivatives and the estimator require conditions that hold uniformly over the parameter space and ensure sufficient curvature. The empirical specification enforces positive definiteness on the observed graph, while the approximation bound additionally depends on the set of eligible pairs and the overall probability of link formation.

Each observed link adds a quadratic contribution to $\mathbf{Q}_0$ and degree-dependent linear terms to $\mathbf{b}$. Coefficient matching in Supplementary Material 6 gives the inverse covariance matrix below. On a graph without cycles, it keeps firm and executive variances fixed as the number of links changes, so we call it the variance-stable precision matrix. We set the linear coefficients on firms’ and executives’ other matching options to zero when estimating the model, so $\mathbf{b} = \mathbf{0}$ and $\mathbf{x} \sim \mathcal{N}(\mathbf{0}, \mathbf{Q}^{-1})$. Nonzero coefficients would instead give the degree-dependent mean $\mathbf{Q}^{-1} \mathbf{b}$.

$$\mathbf{Q} = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}. \quad (5)$$

Here $\mathbf{A}$ records unique firm-executive links, $\mathbf{D}$ records the corresponding node degrees, and $\mathbf{S} = \text{diag}(\sigma_a^{-1} \mathbf{I}_{N_f}, \sigma_z^{-1} \mathbf{I}_{N_m})$ is the inverse scale matrix. The precision matrix is sparse because its off-diagonal entries are nonzero only for linked firms and executives. This sparsity gives the Markov property: conditional on the effects of their directly connected neighbors, effects at more distant nodes are independent.

The AR(1) likelihood has five parameters, $\boldsymbol{\theta} = (\sigma_a, \sigma_z, \rho, \sigma_\varepsilon, \eta)$. On a network without cycles—a forest—firm quality has variance σ_a^2 , executive quality has variance σ_z^2 , every linked firm-executive pair has correlation ρ , and match shocks have variance σ_ε^2 . The parameter η governs the correlation between ordered match shocks at the same firm. Cycles make the corresponding latent second moments depend on the network structure. We therefore use exact graph-implied moments whenever that distinction matters and use variance and correlation terminology in the main text.

6.1.1 The mobility graph and identification from paths. The identification argument is clearest on a forest, where any two connected nodes have a unique path. The correlation between latent effects at graph distance d is then ρ^d . Consider two single-executive outcome edges at distinct firms. If their executives are separated by $h \geq 4$ edges and their firms lie on the connecting path, then

$$\text{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) = \rho^{h-2} (\sigma_a + \rho \sigma_z)^2, \quad (6)$$

so two additional edges multiply the covariance by ρ^2 , independently of the scale parameters, whenever the covariance ratio is defined.

Equation (6) separates the covariance level, which depends on σ_a , σ_z , and ρ , from the decay rate, which depends only on $|\rho|$. We recover the sign of ρ jointly with the variance parameters. Figure 2 evaluates this implication using centered covariance moments from balanced sets of outcome pairs on isolated shortest paths. We report separate series for paths with firm, executive, and mixed endpoints because their covariance levels differ. The figure diagnoses the path-based identification argument; the AR(1) likelihood in the main text instead fits all outcomes jointly.

6.2 From latent effects to observed outcomes

We link latent effects $\mathbf{x}$ to observed outcomes $\mathbf{y}$ by writing equation (1) in matrix form:

$$\mathbf{y} = \mathbf{V} \mathbf{x} + \boldsymbol{\varepsilon}. \quad (7)$$

The $K \times (N_f + N_m)$ design matrix $\mathbf{V}$ maps match-cluster observations to latent types. A single-executive row loads once on its firm and once on its executive. A cluster with q simultaneous executives loads once on the firm and $1/q$ on each executive, so the cluster contributes one outcome and one shock. Let r_g be the chronological rank of cluster g among the observed clusters of its firm. For clusters g and h at the same firm, the corresponding entry of $\mathbf{R}_\eta$ is $\eta^{|r_g - r_h|}$. Entries for clusters at different firms are zero. Distance here counts observed match clusters, not elapsed calendar years. Then

$$\boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \sigma_\varepsilon^2 \mathbf{R}_\eta), \quad \boldsymbol{\Omega} = \mathbf{V} \mathbf{Q}^{-1} \mathbf{V}^\top + \sigma_\varepsilon^2 \mathbf{R}_\eta. \quad (8)$$

This $K \times K$ matrix is dense, making direct inversion infeasible. The within-firm shock blocks are sparse in inverse form, so we can evaluate the likelihood in node space.

6.3 Maximum likelihood and computation

The log-likelihood is

$$\ell(\theta) = -\frac{K}{2} \ln(2\pi) - \frac{1}{2} \ln \det \Omega - \frac{1}{2} \mathbf{y}^\top \Omega^{-1} \mathbf{y}. \quad (9)$$

Here $\theta = (\sigma_a, \sigma_z, \rho, \sigma_\varepsilon, \eta)$. The Woodbury identity replaces operations on Ω with operations on $\mathbf{P} = \mathbf{Q} + \mathbf{V}^\top (\sigma_\varepsilon^2 \mathbf{R}_\eta)^{-1} \mathbf{V}$, an $(N_f + N_m) \times (N_f + N_m)$ matrix. For single-executive observations, $\mathbf{V}^\top (\sigma_\varepsilon^2 \mathbf{R}_\eta)^{-1} \mathbf{V}$ adds firm-executive entries and, through the AR(1) blocks, entries between adjacent match clusters within firms. Grouped teams also add small executive-executive blocks. Thus $\mathbf{P}$ has the union of prior-graph, within-firm, and within-team sparsity and remains sparse even though Ω is dense. Exact sparse Cholesky factorizations supply the log-determinant and quadratic form. The preferred fit contains 2,340,791 latent nodes and converges with ExactCholesky under finite-difference optimization, a 1,500-iteration limit, and relative tolerance 10^{-7} .

6.4 Precision weighting and graph restrictions

Equation (5) adjusts each diagonal entry for the degree of its node. Under the plain precision $\mathbf{S}(\mathbf{D} - \rho \mathbf{A})\mathbf{S}$, marginal correlations vary with degree. Under the variance-stable precision, every linked pair on a forest has correlation ρ , while firms and executives have marginal variances σ_a^2 and σ_z^2 . This identity interprets the parameters; it does not test the link-formation approximation.

Several features distinguish the fitted graph and outcome design from the forest example.

Cycles. Alternative paths change marginal variances and covariances, so ρ need not equal the average correlation of linked effects. We calculate the decomposition from $\mathbf{Q}^{-1}$ on the fitted graph rather than from forest scale formulas.

Positive definiteness. Consider walks that never immediately reverse the preceding edge. The non-backtracking spectral radius, λ_{NB} , is a graph statistic calculated from these walks. Equation (5) is positive definite only when $|\rho| \lambda_{NB} < 1$. The mathematical ceiling is $\rho_{PD} = 1/\lambda_{NB}$; the optimizer stays inside it using the conservative box $\rho_{box} = \min\{0.95, 0.98/\lambda_{NB}\}$. We compute λ_{NB} from the Ihara-Bass companion matrix rather than constructing the larger directed-edge matrix. The bound depends only on the graph, not on outcomes.

6.4.1 Pruning the mobility graph. We prune cycle-closing edges while retaining every node and connected component. The algorithm first protects a spanning forest: a set of edges that connects every node within each component without creating a cycle. It then removes unprotected edges that contribute to the leading eigenvector used to calculate the non-backtracking spectral radius and recomputes the radius as the graph changes. The graph is fixed before fitting: the spectral criterion depends on the mobility matrix, not on revenue outcomes. This rule enforces computational feasibility on the fitted graph. The stronger approximation bound in Supplementary Material 6 additionally requires feasibility after adding each potential edge in the eligible set; our pruning criterion does not impose that condition.

Missing outcomes. Links without observed firm performance still connect firms and executives. They enter the graph and its precision matrix but add no row to the outcome vector. They therefore

affect the likelihood indirectly through the covariance of observed outcomes. The graph used for that covariance must also be the graph used to calculate the spectral feasibility bound.

Multiple simultaneous executives. A match cluster is a firm interval over which the set of co-present executives is constant. When a cluster has several executives, its design row loads once on the firm effect and on the average of the executives' effects. The cluster therefore has one shock term, rather than crediting every executive with the full firm outcome. Supplementary Material 3 derives this grouped observation; Supplementary Material 4 explains its effect on the sparse likelihood calculation.

6.5 Observables

Observable characteristics enter as

$$y_{im} = \mathbf{w}_{im}^\top \boldsymbol{\beta} + a_i + z_m + \varepsilon_{im},$$

where $\mathbf{w}_{im}$ contains the included observable controls. Observables shift the mean without changing the maintained covariance structure. The preferred specification uses log real revenue demeaned by calendar year before match-cluster grouping and does not include industry effects or degree-dependent mean coefficients. For the graph-coverage robustness check, we instead demean outcomes by year and industry and adjust for the numbers of graph links of the firm and executive.

6.6 Data sources and construction

We combine two administrative sources covering Hungarian firms [20, 21]. The corporate-court registry records all executives authorized to represent a company, including chief executives and other officers with signatory authority [11]. Annual balance sheets for all firms with double-entry bookkeeping provide revenue, employment, sector, and ownership. Together, the sources cover more than one million firms.

We identify records that refer to the same executive across firms and time, a process known as entity resolution. Numerical identifiers provide direct links from 2013 onward. In earlier years, we construct person identifiers from names, addresses, mothers' names, and birthdates. Link quality improves after mothers' names enter in 1999 and birthdates in 2010; names and addresses provide the links before then.

We identify chief executives among registered representatives in three steps. First, we use managing-director titles where available. Second, we classify all representatives as chief executives when a firm has no more than three. Third, for firms with more representatives, we use continuity with previously identified chief executives.

We apply five sample restrictions. First, we exclude mining and finance because they use specialized accounting. Second, we drop firms that ever have more than two simultaneous chief executives. Third, we remove firms with more than ten executive changes, which are likely to contain classification error. Fourth, we exclude state-owned enterprises. Fifth, we exclude public firms, joint-stock companies, and cooperatives because their governance differs from that of private corporations. We retain private nonemployer firms because single-person companies form a substantial part of the

Hungarian corporate sector. The resulting sample therefore contains private, non-state Hungarian corporations with double-entry accounts outside mining and finance that have no more than two simultaneous chief executives and no more than ten executive changes. It includes private nonemployers but excludes public firms, joint-stock companies, and cooperatives.

The outcome is annual log real firm revenue, deflated by standard price indices and demeaned by calendar year. We average these annual residuals within each interval over which a firm's executive set is constant. Each resulting match cluster contributes one grouped outcome to the likelihood. When several executives serve simultaneously, the outcome loads on the average of their effects.

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Author contributions. M.K. conceived the idea; M.K. and K.O. designed the study; M.K. and U.W. prepared the data; U.W. developed computational methods and performed data analysis; M.K., K.O., and U.W. wrote the paper.

Competing interests. The authors declare no competing interests.

Data, code, and materials availability. The restricted-use Cégjegyzék LTS and Mérleg LTS source data are distributed by HUN-REN KRTK and published by Opten Zrt. [20, 21]. Opten has authorized release of the study's full pseudonymized edge list, containing pseudonymous firm identifiers, pseudonymous executive identifiers, and incrementing match identifiers. Before publication, we will deposit that edge list, all data-cleaning code, all analysis code, and documentation in a public DOI-minting repository such as Zenodo or Figshare. No physical materials were generated in this work.

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SUPPLEMENTARY MATERIALS

Materials and Methods
Supplementary Materials 1–6
References

SUPPLEMENTARY MATERIAL 1: COVARIANCE ALONG NETWORK PATHS

Under the maintained latent GMRF model, covariance decay along network paths identifies sorting separately from the scales of firm and executive heterogeneity. We derive the result on a forest, where each pair of connected nodes has a unique path and covariance has a closed form. The likelihood uses the declared estimation graph, including its retained cycles and all connected components; the forest supplies the identification argument.

Let $G = (V, E)$ be the undirected bipartite graph of firms, executives, and their observed links. We use $d(u, v)$ for graph distance, the number of edges in the shortest path between nodes u and v . A path alternates between firms and executives. Distances between two firms or two executives are therefore even, whereas distances between a firm and an executive are odd.

For an edge (i, m) , the outcome equation is

$$y_{im} = a_i + z_m + \varepsilon_{im}, \quad \varepsilon_{im} \sim \mathcal{N}(0, \sigma_\varepsilon^2), \quad (10)$$

where a_i is firm quality, z_m is executive quality, and the match-specific error is independent of all latent types. In the preferred specification, ordered match clusters at the same firm have AR(1) correlation; clusters at different firms have independent errors. The covariance formulas below apply to outcome pairs at different firms, so their error covariance is zero. A same-firm pair additionally includes the within-firm error covariance. Equation (10) describes a match with one executive. Supplementary Material 3 explains how the design row changes when several executives are present simultaneously.

Under the variance-stable parameterization, the precision is

$$\mathbf{Q}(\rho) = \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S},$$

$$\mathbf{S} = \text{diag} \left(\sigma_a^{-1} \mathbf{I}_{N_f}, \sigma_z^{-1} \mathbf{I}_{N_m} \right). \quad (11)$$

On a forest, inversion of equation (11) gives

$$\text{Cov}(a_i, a_j) = \sigma_a^2 \rho^{d(i,j)}, \quad (12)$$

$$\text{Cov}(z_m, z_n) = \sigma_z^2 \rho^{d(m,n)}, \quad (13)$$

$$\text{Cov}(a_i, z_m) = \sigma_a \sigma_z \rho^{d(i,m)}. \quad (14)$$

The scale parameter at each endpoint determines the covariance level, while ρ is multiplied once for each edge on the unique path. In particular, firms have marginal variance σ_a^2 , executives have marginal variance σ_z^2 , and a linked firm–executive pair has correlation ρ . The normalization outside the middle matrix is what gives the two scale parameters this marginal-variance interpretation on a forest.

Intuitively, each edge transmits a fraction ρ of standardized type. Moving over d edges therefore multiplies covariance by ρ^d ; the endpoint scales convert it back to the units of a and z .

For two outcome edges (i, m) and (j, n) at different firms, the match errors are independent. Their covariance is therefore the sum of four latent covariances:

$$\begin{aligned} \text{Cov}(y_{im}, y_{jn}) = & \sigma_a^2 \rho^{d(i,j)} + \sigma_a \sigma_z \rho^{d(i,n)} \\ & + \sigma_a \sigma_z \rho^{d(m,j)} + \sigma_z^2 \rho^{d(m,n)}. \end{aligned} \quad (15)$$

The four terms correspond, in order, to firm–firm, firm–executive, executive–firm, and executive–executive covariance. Equation (15) is the general forest formula. A scalar decay law additionally requires us to fix how the two observed edges are oriented relative to their connecting path.

Consider first the executive–executive configuration in the main text,

$$m_1 - i_1 - \cdots - i_2 - m_2,$$

where the executive endpoints are $h = d(m_1, m_2)$ edges apart and both observed firm endpoints lie on the connecting path. The four distances in equation (15) are then $h - 2$, $h - 1$, $h - 1$, and h . For distinct endpoint firms, $h \geq 4$. Collecting terms gives

$$C_h^{EE, \text{in}} = \text{Cov}(y_{i_1 m_1}, y_{i_2 m_2}) = \rho^{h-2} (\sigma_a + \rho \sigma_z)^2. \quad (16)$$

For the analogous firm–firm configuration,

$$i_1 - m_1 - \cdots - m_2 - i_2,$$

where $h = d(i_1, i_2)$ and the executive endpoints lie on the path, the expression is

$$C_h^{FF, \text{in}} = \rho^{h-2} (\sigma_z + \rho \sigma_a)^2. \quad (17)$$

For the oriented firm–executive configuration

$$i_1 - m_1 - \cdots - i_2 - m_2,$$

with $h = d(i_1, m_2) \geq 3$, the expression is

$$C_h^{FE} = \rho^{h-2} (\sigma_a + \rho \sigma_z)(\sigma_z + \rho \sigma_a). \quad (18)$$

If an observed edge branches away from the connecting path, the four shifted distances differ from those above, but equation (15) remains valid. This is why the orientation must be held fixed when comparing covariance at different distances.

Within any fixed orientation with nonzero covariance, increasing the path length by two multiplies the covariance by

$$\frac{C_{h+2}}{C_h} = \rho^2. \quad (19)$$

The covariance level depends on σ_a , σ_z , and ρ , whereas the likeness decay ratio depends only on ρ^2 . The decay rate therefore identifies $|\rho|$. Odd-distance moments contain information about its sign, which is estimated jointly with the scale parameters in the Gaussian likelihood. Their sign alone need not equal that of ρ when the endpoint scales differ and $\rho < 0$. Match-specific noise affects each outcome's variance and covariance between observations at the same firm. Under the maintained error model it does not enter covariance between outcomes at different firms.

Cycles make covariance depend on the surrounding topology, not only on shortest distance. The likelihood therefore uses the relevant entry of $\mathbf{Q}^{-1}$ on the actual graph. The forest formulas remain the economic intuition: longer network paths reveal how quickly dependence between firm and executive quality decays.

Empirical covariance-decay diagnostic

Figure 2 in the main text constructs empirical counterparts to the oriented moments above. Starting from observed match edges, we extend non-backtracking walks in both directions on the fitted prior graph. For each distance h , we retain the two endpoint outcomes only when the intervening segment is a shortest path and sufficiently isolated from alternative paths. A common-path design

balances the comparison across the required distances and endpoint orientations.

Even-distance paths produce separate firm–firm and executive–executive series, while odd-distance paths produce the mixed firm–executive series. The outcomes are measured in their original units after calendar-year demeaning. Within each endpoint orientation and distance cell, we center the outcomes at the left and right endpoints separately. The empirical moment is the average product of these centered outcomes. Its standard error is the standard deviation of the products divided by the square root of the number of retained pairs; Figure 2 reports 95% confidence intervals.

The dashed maximum-likelihood curves evaluate the orientation-specific forest formulas at the independent-error estimates. They are not refitted to the selected paths, so both their level and slope are model-implied. Observed covariance declines more slowly than these curves in several cells. The figure therefore shows where the independent-error specification misses the selected path moments.

A separate minimum-distance calculation approximates these selected moments. It is not drawn in Figure 2. Table 3 reports a different minimum-distance estimate, based on the comparison graph and excluding the within-firm executive–executive cell.

SUPPLEMENTARY MATERIAL 2: SPECTRAL FEASIBILITY AND EDGE PRUNING

Cycles also restrict which correlations the Gaussian field can support. The precision in equation (11) must be positive definite. Because the diagonal scale matrix S is positive definite, feasibility is determined by the middle matrix

$$H(\rho) = (1 - \rho^2)I + \rho^2 D - \rho A. \quad (20)$$

For the variance-stable construction, its positive-definiteness condition can be written as

$$|\rho| \lambda_{NB} < 1, \quad (21)$$

where λ_{NB} is the non-backtracking spectral radius of the graph. The restriction is known before observing outcomes because it depends only on the graph and ρ . A forest has no closed non-backtracking walk and hence $\lambda_{NB} = 0$, so every $|\rho| < 1$ is feasible.

The non-backtracking matrix is indexed by directed edges. Its (e, f) entry is one when edge f can follow edge e without immediately reversing direction. Its spectral radius measures how quickly such walks multiply. Dense cyclic regions raise λ_{NB} ; pendant trees do not affect it.

We calculate λ_{NB} using the sparse Ihara–Bass companion matrix

$$C = \begin{pmatrix} A & I - D \\ I & 0 \end{pmatrix}, \quad (22)$$

rather than explicitly forming the larger directed-edge Hashimoto matrix. This linearization preserves the relevant non-backtracking eigenvalues while using only $O(|V| + |E|)$ nonzero entries. Before computing the leading eigenvalue, we remove degree-zero and degree-one nodes recursively and operate on the 2-core. This reduction changes computation but not λ_{NB} .

The raw empirical graph has $\lambda_{NB} = 15.26$. The mathematical positive-definiteness ceiling is therefore

$$\rho_{PD} = 1/\lambda_{NB} \approx 0.066. \quad (23)$$

Conventional restrictions on component size or minimum degree retain the dense cyclic core that generates this value. At the same time, they remove peripheral firms and executives with fewer mobility links, shifting the economic population toward more connected firms and careers without materially relaxing the ceiling. They therefore reduce coverage through economic selection while leaving the binding region in place.

We instead target the mechanism that tightens the ceiling by removing selected cycle-closing edges. The pruning algorithm first protects a spanning forest. It then removes unprotected edges that contribute to the leading non-backtracking eigenvector, recomputing the radius as the graph changes. Protecting the forest preserves every node and connected component; only redundant paths within cyclic regions are removed. The preferred match-cluster graph uses the target $\lambda_{NB} < 1.5$. Spanning-forest estimates provide the limiting comparison in which all cycle-closing edges are removed.

We distinguish the mathematical ceiling from the bound used by the optimizer. The pruning stage records the achieved radius and sets the conservative box

$$\rho_{\text{box}} = \min\{0.95, 0.98/\lambda_{NB}\}. \quad (24)$$

The factor 0.98 keeps numerical optimization away from the singular boundary, and the cap at 0.95 keeps it away from $|\rho| = 1$ even when the graph is a forest. The estimator reads this graph-specific bound from the pruning output. It does not recompute the radius or apply a common bound to graphs with different cyclic structure.

Positive definiteness ensures that a given correlation defines a proper Gaussian field on the retained graph. Accuracy relative to the complete link-formation model additionally depends on the risk set, link intensity, moment envelopes, and the augmented-graph margin in Supplementary Material 6. Our pruning rule controls the radius of the retained graph, not this augmented radius. Comparisons along a pruning path must hold the error process, outcome construction, and fitted parameter values fixed.

SUPPLEMENTARY MATERIAL 3: MATCH OBSERVATIONS AND SIMULTANEOUS EXECUTIVES

The administrative data can list several executives for the same firm over the same interval. Treating each executive–firm row as an independent outcome would repeat the same firm performance and attach a separate error term to each copy. It would therefore give mechanically greater weight to firms with co-executives and overstate the amount of independent information.

We instead define a match cluster as a firm interval over which the set of co-present executives is constant. If cluster g contains firm i and the set M_g of q_g executives, its observation equation is

$$y_g = a_i + \frac{1}{q_g} \sum_{m \in M_g} z_m + \varepsilon_g. \quad (25)$$

The firm receives one loading, the executive input is their average, and the cluster receives one match disturbance. The construction allows executives to have different types but imposes symmetric loadings because the data do not allocate firm revenue among team members. The single-executive equation in Supplementary Material 1 is the special case $q_g = 1$.

The prior graph and the outcome design serve different purposes. Every retained firm–executive relation enters $\mathbf{A}$ and $\mathbf{D}$, including a link in a cluster without a finite outcome. Only clusters with a finite grouped outcome enter $\mathbf{y}$ and $\mathbf{V}$. Such graph-only links can still connect observed outcomes through the latent field, but they do not add a row to the likelihood. In the preferred match-cluster fit, the prior graph contains 191,659 graph-only firm–executive links, while the likelihood contains $K = 1,666,857$ grouped outcomes. These counts refer to distinct objects and should not be expected to coincide. Supplementary Material 4 derives the likelihood and the resulting sparsity pattern.

SUPPLEMENTARY MATERIAL 4: EXACT LIKELIHOOD AND COMPUTATION

Sparsity makes the Gaussian marginal likelihood feasible even with millions of latent effects. Rather than estimate a fixed effect for each firm and executive, we integrate them out. The preferred AR(1) specification has five covariance and error parameters. We derive the likelihood with a general mean; the estimates impose the zero-mean restriction described below.

Stack the latent effects as $\mathbf{x} = (\mathbf{a}^\top, \mathbf{z}^\top)^\top$. For K observed match outcomes, write the model as

$$\mathbf{y} = \mathbf{V}\mathbf{x} + \boldsymbol{\varepsilon}, \quad \boldsymbol{\varepsilon} \sim \mathcal{N}(\mathbf{0}, \mathbf{E}_\eta), \quad \mathbf{E}_\eta = \sigma_\varepsilon^2 \mathbf{R}_\eta. \quad (26)$$

For two ordered match clusters g and h at the same firm, $(\mathbf{R}_\eta)_{gh} = \eta^{|r_g - r_h|}$; clusters at different firms have zero error covariance. The matrix $\mathbf{V}$ contains the loadings defined in equation (25); each row is sparse and corresponds to one match disturbance.

Write the prior as

$$\mathbf{x} \sim \mathcal{N}(\boldsymbol{\mu}, \mathbf{Q}^{-1}), \quad \boldsymbol{\mu} = \mathbf{Q}^{-1}\mathbf{b}, \quad (27)$$

where Supplementary Material 6 derives the degree-dependent linear term $\mathbf{b}$ from link formation. The preferred estimates set $\mathbf{b} = \mathbf{0}$. Integrating out $\mathbf{x}$ gives

$$\mathbf{y} \sim \mathcal{N}(\mathbf{V}\boldsymbol{\mu}, \boldsymbol{\Omega}), \quad \boldsymbol{\Omega} = \mathbf{V}\mathbf{Q}^{-1}\mathbf{V}^\top + \mathbf{E}_\eta. \quad (28)$$

The first term transmits dependence through the mobility graph. Even if two rows of $\mathbf{V}$ share no node, their outcomes can be correlated because the corresponding latent types are connected through $\mathbf{Q}^{-1}$. For this reason, $\boldsymbol{\Omega}$ is dense within a connected component. Forming or factorizing it directly would discard the computational advantage of the sparse prior.

We instead work with the conditional distribution of $\mathbf{x}$ given $\mathbf{y}$. Combining the quadratic forms in the Gaussian prior and Gaussian observation density gives the posterior precision

$$\mathbf{P} = \mathbf{Q} + \mathbf{V}^\top \mathbf{E}_\eta^{-1} \mathbf{V}. \quad (29)$$

The corresponding posterior mean is

$$\mathbb{E}[\mathbf{x} | \mathbf{y}] = \boldsymbol{\mu} + \mathbf{P}^{-1} \mathbf{V}^\top \mathbf{E}_\eta^{-1} (\mathbf{y} - \mathbf{V}\boldsymbol{\mu}). \quad (30)$$

Thus $\mathbf{P}$ is both the precision after observing outcomes and the sparse matrix that makes the marginal likelihood computable.

Let $\tilde{\mathbf{y}} = \mathbf{y} - \mathbf{V}\boldsymbol{\mu}$. The Gaussian log-likelihood requires $\log \det \boldsymbol{\Omega}$ and $\tilde{\mathbf{y}}^\top \boldsymbol{\Omega}^{-1} \tilde{\mathbf{y}}$. The matrix determinant lemma and Woodbury identity convert both objects from outcome space to node space:

$$\log \det \boldsymbol{\Omega} = \log \det \mathbf{E}_\eta + \log \det \mathbf{P} - \log \det \mathbf{Q}, \quad (31)$$

$$\tilde{\mathbf{y}}^\top \boldsymbol{\Omega}^{-1} \tilde{\mathbf{y}} = \tilde{\mathbf{y}}^\top \mathbf{E}_\eta^{-1} \tilde{\mathbf{y}} - (\mathbf{V}^\top \mathbf{E}_\eta^{-1} \tilde{\mathbf{y}})^\top \mathbf{P}^{-1} (\mathbf{V}^\top \mathbf{E}_\eta^{-1} \tilde{\mathbf{y}}). \quad (32)$$

Substituting equation (32) into the Gaussian density gives the exact marginal log-likelihood without constructing $\boldsymbol{\Omega}$. All factorizations are performed in the $N_f + N_m$ dimensional node space rather than the K dimensional outcome space.

Sparsity is preserved, although correlated errors and grouped observations enlarge the pattern. The inverse AR(1) blocks connect adjacent ordered clusters within firms. With one executive per observation, $\mathbf{V}^\top \mathbf{E}_\eta^{-1} \mathbf{V}$ also has firm–executive off-diagonals on observed links. Simultaneous executives add executive–executive off-diagonals within match clusters. Consequently, $\mathbf{P}$ has the union of the prior-graph, within-firm, and within-cluster patterns. It remains sparse even though $\boldsymbol{\Omega}$ is dense.

We evaluate the preferred fit with exact sparse Cholesky factorizations, finite-difference derivatives, a 1,500-iteration limit, and relative tolerance 10^{-7} . For a given mean specification, Cholesky factors supply the log determinants and the solve in equation (32). These Gaussian identities are exact conditional on the specified $\mathbf{Q}$, $\mathbf{V}$, and $\mathbf{E}_\eta$. Stochastic trace calculations in the variance decomposition use a fixed random seed; the exact likelihood does not depend on that seed.

When Cholesky fill-in is prohibitive, we combine a Hutchinson trace estimator with Lanczos quadrature. For independent Rademacher probes $\mathbf{r}_b$,

$$\log \det \mathbf{P} = \text{tr}(\log \mathbf{P}) \approx \frac{1}{B} \sum_{b=1}^B \mathbf{r}_b^\top \log(\mathbf{P}) \mathbf{r}_b, \quad (33)$$

and Lanczos quadrature approximates each probe quadratic form using only sparse matrix–vector products. The same method applies to $\log \det \mathbf{Q}$. This stochastic route provides a scalable alternative when Cholesky fill-in is prohibitive; the preferred likelihood uses exact Cholesky.

The decomposition concerns a target distribution of outcomes on the stated graph, not posterior uncertainty about a particular node. It therefore uses the prior covariance $\mathbf{Q}^{-1}$, not the posterior covariance $\mathbf{P}^{-1}$. For a design row $\mathbf{v}_k$, the latent contribution is

$$\text{Var}(\mathbf{v}_k^\top \mathbf{x}) = \mathbf{v}_k^\top \mathbf{Q}^{-1} \mathbf{v}_k. \quad (34)$$

Expanding this quadratic form separates firm variance, executive variance, and twice the firm–executive covariance. We average these terms over an edge target using common stochastic probes. The error contribution is defined on the same edge target and is not, in general, the square of the row-level scale parameter σ_ε . On a cyclic graph the latent terms are computed from $\mathbf{Q}^{-1}$ rather than substituted from the forest formulas.

SUPPLEMENTARY MATERIAL 5: FIXED-EFFECTS COMPARISON

Leave-one-out restrictions and leverage

Table S1 compares the GMRF with fixed-effects and leave-out estimators in the dense Veneto Employer History [23]. The naive

plug-in correlation is -0.086 ; the homoscedastic correction gives 0.105 , and leave-one-out gives 0.155 . On the same $56,044$ spells, the degree-weighted GMRF gives 0.160 . The dense worker sample therefore shows that the GMRF and leave-out estimators yield similar sorting estimates when mobility is extensive.²

Panel B compares estimators under independent match errors on the sparse Hungarian executive network. On the common subgraph, the naive fixed-effects correlation is -0.594 , leave-one-match-out gives -0.162 , and the GMRF gives $+0.351$. The three estimates use the same $91,085$ matched pairs, so sample composition does not explain their disagreement. This comparison isolates estimator differences under independent errors; the preferred AR(1) specification instead models serial correlation between match clusters within firms [23].

Leave-out admissibility can sharply restrict a sparse graph [2, 8, 23]. The match-level procedure first removes bridges and retains the largest remaining block; the common graph in Table S1 imposes a further restriction so that all estimators use the same observations. This operation differs from repeated articulation-point pruning and from deleting all observations at one firm. The leave-out unit determines both the admissible sample and the error dependence that a correction allows [32].

Leave-one-out admissibility selects a small, mobility-intensive part of the executive network. The random-effects likelihood does not require this graph restriction. To see why fixed effects do, let $\mathbf{X}$ denote the two-way fixed-effects design on the connected comparison sample. The leverage of observation k is

$$h_k = \mathbf{x}_k^\top (\mathbf{X}^\top \mathbf{X})^+ \mathbf{x}_k, \quad (35)$$

where $+$ denotes a generalized inverse after the usual location normalization. A leave-one-out correction requires $h_k < 1$ for every retained observation. If $h_k = 1$, deleting observation k removes all identifying variation for part of the fixed-effects design, so its leave-one-out residual is undefined.

In a bipartite matched-pair design, a bridge is an edge whose deletion disconnects the graph. Such an edge has leverage one. Sparse executive careers produce many bridges: an executive observed at only one firm, a peripheral firm, or a thin connection between two otherwise dense regions can all create one. The problem is therefore economic as well as numerical. Admissibility selects executives with mobility and firms located in redundant mobility paths.

Our match-level comparison first collapses the data to one observation per unique firm-executive pair. It then drops all bridges in a single pass and retains the largest 2-edge-connected block returned by the leave-out package. This is the sample on which our leave-one-match-out correction is defined. It should be distinguished from implementations that repeatedly remove articulation points or apply a different leave-out unit. The full funnel is therefore: finite outcomes, unique matched pairs, non-bridge pairs, and the largest admissible block.

For the same-graph comparison, spectral pruning is applied so that the GMRF and leave-one-out estimators use identical matched pairs. On this common graph the leave-out design has $91,085$ observations, $38,701$ firms, and $35,324$ executives. Its leverage sum is

²The worker comparison uses wages, for which firm-worker covariance can combine production complementarities with rent sharing [9, 10, 27]. The executive application instead uses firm revenue, so the two outcomes measure different economic objects.

$74,024$, equal to the rank $N - 1$ of the connected two-way design; its maximum leverage is 0.973684 , strictly below one. The norm of $\mathbf{X}^\top \hat{\boldsymbol{\epsilon}}$ is 1.867×10^{-11} , which verifies the fixed-effects normal equations. These diagnostics verify that the correction is numerically defined on the selected graph. Its economic population is the mobility-intensive subset that satisfies leave-out admissibility.

The estimands also differ. Two-way fixed effects estimate realized firm and executive effects on the selected connected graph; under the required leave-out error assumptions, the correction removes finite-sample bias in their quadratic variance components. It does not generally remove bias in a nonlinear correlation ratio. The GMRF estimates a distribution of latent effects and reports its variance decomposition on a stated graph. On a forest, ρ is exactly the linked-type correlation; on a cyclic graph the corresponding marginal moment is computed from $\mathbf{Q}^{-1}$.

The dense worker comparison therefore provides a benchmark for second moments on one sample. The executive comparison shows that fixed-effects and GMRF estimators disagree even on identical observations in a selected, mobility-intensive subgraph. Founder exclusions, alternative mean controls, and firm-cluster leave-out address separate sources of sensitivity.

SUPPLEMENTARY MATERIAL 6: A LINK-FORMATION INTERPRETATION OF THE PRECISION AND MEAN

We start from independent population types, selective link formation, and outcomes that depend on the selected types. The complete link pattern weights the population density by both formed-link and absent-link probabilities. Replacing formed-link probabilities by exponential kernels and omitting the shared logit denominators yields the variance-stable Gaussian field under coefficient matching. Proposition 1 bounds the resulting error for a fixed, finite risk set under an augmented-graph spectral margin and sparse-link moment conditions. Applying the bound to an empirical graph requires these conditions to hold for its risk set and link intensity.

This is a link-selection model rather than a competitive equilibrium model of matching. The empirical GMRF does not specify a risk set, capacity constraints, transfers, or the joint determination of outside options. We introduce a finite risk set below to compare the positive-link model with a complete logistic link pattern.

Independent population types

Before links form, let firm type a and executive type z be independent:

$$\begin{pmatrix} a \\ z \end{pmatrix} \sim \mathcal{N} \left[\begin{pmatrix} 0 \\ 0 \end{pmatrix}, \begin{pmatrix} \omega_a^2 & 0 \\ 0 & \omega_z^2 \end{pmatrix} \right]. \quad (36)$$

Independence is the identifying benchmark. Any covariance between a and z among linked pairs must arise from link selection rather than from covariance in the population from which firms and executives are drawn.

For a graph with N_f firms and N_m executives, stack all types in $\mathbf{x} = (\mathbf{a}^\top, \mathbf{z}^\top)^\top$. The population prior is

$$p_0(\mathbf{x}) \propto \exp \left(-\frac{1}{2} \mathbf{x}^\top \mathbf{Q}_0 \mathbf{x} \right), \quad \mathbf{Q}_0 = \text{diag}(\omega_a^{-2} \mathbf{I}_{N_f}, \omega_z^{-2} \mathbf{I}_{N_m}). \quad (37)$$

Table S1: Estimator comparison on dense and sparse mobility networks. Each column reports an estimator fit on the stated comparison sample.

Statistic	naive FE	homosced.	leave-out	weighted GMRF
<i>Panel A. Dense worker–firm replication data</i>				
Firm–worker corr.	−0.086	0.105	0.155	0.160
Job spells	56,044	56,044	56,044	56,044
<i>Panel B. Hungarian executive data, common subgraph</i>				
Firm variance	8.216	?	4.223	3.681
Executive variance	4.408	?	0.435	0.023
Firm–executive corr.	−0.594	?	−0.162	+0.351
Match-noise variance	—	?	1.495	1.656
Match observations	91,085	?	91,085	91,085

Notes: Panel A uses the leave-one-worker-out connected set from the worker–firm replication data. Panel B uses the common Hungarian executive subgraph with one outcome per unique firm–executive pair. The fixed-effects columns report moments of realized effects; the weighted GMRF column reports distributional variance components and its sorting parameter. Panel B assumes independent errors across matched pairs; the preferred match-cluster specification models within-firm errors as AR(1).

The diagonal precision is important. Link selection will add both positive and negative curvature to this prior; only their sum must define a proper Gaussian distribution.

A sparse link probability

Let L_{im} denote a link between firm i and executive m . We write the systematic index as match surplus net of type-specific outside options,

$$\eta(a_i, z_m) = \kappa + \frac{1}{\tau} \left[a_i z_m - \alpha_1 a_i - \zeta_1 z_m - \frac{\alpha_2}{2} a_i^2 - \frac{\zeta_2}{2} z_m^2 \right], \quad (38)$$

where $\tau > 0$ scales match-specific frictions. The cross derivative of the surplus term is positive, so higher firm and executive types are complements. The linear and quadratic terms allow their outside options to vary with type.

Suppose the link probability is logistic,

$$\Pr(L_{im} = 1 \mid a_i, z_m) = \frac{\exp[\eta(a_i, z_m)]}{1 + \exp[\eta(a_i, z_m)]}. \quad (39)$$

Executive–firm networks are sparse: among all potential pairs, a link is a rare event. When $\exp(\eta)$ is small,

$$\Pr(L_{im} = 1 \mid a_i, z_m) \simeq \exp[\eta(a_i, z_m)]. \quad (40)$$

Equation (40) approximates each formed-link probability. A complete link-pattern likelihood also contains the probabilities of absent links. Dropping those terms is a separate step. Together these steps produce the positive-link kernel below; they do not give the exact likelihood of a complete logistic network. Individually unlikely links need not make the aggregate contribution of nonlinks small. Proposition 1 makes this distinction explicit on a finite risk set.

Positive-link tilt from the observed graph

Let E contain the unique firm–executive links in the realized graph. This is the same binary-link convention used by the prior adjacency matrix in the estimator: repeated observations of one pair do not add

repeated prior links. Under the sparse approximation, the positive-link tilted density of all node types is

$$p_E^+(\mathbf{x}) \propto p_0(\mathbf{x}) \prod_{(i,m) \in E} \exp[\eta(a_i, z_m)] \quad (41)$$

$$\propto \exp\left(-\frac{1}{2} \mathbf{x}^\top \mathbf{Q} \mathbf{x} + \mathbf{b}^\top \mathbf{x}\right). \quad (42)$$

We can read the tilted precision and linear term directly from the exponent. Each observed link contributes

$$\Lambda_{im} = \frac{1}{\tau} \begin{pmatrix} \alpha_2 & -1 \\ -1 & \zeta_2 \end{pmatrix} \quad (43)$$

to the rows and columns for (a_i, z_m) . Summing over links puts $\alpha_2 d_i / \tau$ on firm i 's diagonal, $\zeta_2 d_m / \tau$ on executive m 's diagonal, and $-1/\tau$ in each linked firm–executive position. Hence

$$\mathbf{Q} = \mathbf{Q}_0 + \frac{1}{\tau} (\mathbf{C} \mathbf{D} - \mathbf{A}), \quad \mathbf{C} = \text{diag}(\alpha_2 \mathbf{I}_{N_f}, \zeta_2 \mathbf{I}_{N_m}). \quad (44)$$

The linear terms also accumulate with degree:

$$\mathbf{b} = -\frac{1}{\tau} \begin{pmatrix} \alpha_1 \mathbf{d}_f \\ \zeta_1 \mathbf{d}_m \end{pmatrix}, \quad \mathbb{E}_E^+[\mathbf{x}] = \mathbf{Q}^{-1} \mathbf{b}. \quad (45)$$

Thus quadratic outside options govern covariance under the positive-link tilt, while linear outside options generate degree-dependent means. The two ratios α_1/τ and ζ_1/τ could be estimated through the mean of observed outcomes in an extended specification. This derivation does not show that the reported covariance estimates include that extension.

Exact variance-stable parameterization

Equation (44) becomes exactly the variance-stable precision in equation (11) under the following coefficient matching:

$$\begin{aligned} Q_{0,a} &= \frac{1}{\sigma_a^2}, & Q_{0,z} &= \frac{1}{\sigma_z^2}, \\ \frac{\alpha_2}{\tau} &= \frac{\rho^2}{\sigma_a^2(1-\rho^2)}, & \frac{\zeta_2}{\tau} &= \frac{\rho^2}{\sigma_z^2(1-\rho^2)}, & \frac{1}{\tau} &= \frac{\rho}{\sigma_a \sigma_z(1-\rho^2)}. \end{aligned} \quad (46)$$

The first line fixes the independent population variances at

$$\omega_a^2 = \sigma_a^2, \quad \omega_z^2 = \sigma_z^2. \quad (48)$$

The second line matches the degree and adjacency coefficients. With the complementarity coefficient on $a_i z_m$ normalized to one, it implies

$$\tau = \frac{\sigma_a \sigma_z (1 - \rho^2)}{\rho}, \quad \alpha_2 = \rho \frac{\sigma_z}{\sigma_a}, \quad \zeta_2 = \rho \frac{\sigma_a}{\sigma_z}. \quad (49)$$

This interpretation applies directly to positive assortative matching, $\rho > 0$. A negative ρ would require a negative complementarity coefficient rather than a negative friction parameter τ .

Substituting equation (47) into equation (44) gives

$$\begin{aligned} \mathbf{Q} &= \text{diag} \left(\frac{1}{\sigma_a^2} \mathbf{I}_{N_f}, \frac{1}{\sigma_z^2} \mathbf{I}_{N_m} \right) \\ &+ \text{diag} \left(\frac{\rho^2}{\sigma_a^2 (1 - \rho^2)} \mathbf{D}_f, \frac{\rho^2}{\sigma_z^2 (1 - \rho^2)} \mathbf{D}_m \right) - \frac{\rho}{\sigma_a \sigma_z (1 - \rho^2)} \mathbf{A} \\ &= \frac{1}{1 - \rho^2} \mathbf{S} \left[(1 - \rho^2) \mathbf{I} + \rho^2 \mathbf{D} - \rho \mathbf{A} \right] \mathbf{S}. \end{aligned} \quad (50)$$

This equality holds for the positive-link quadratic kernel, not for the exact logistic selected-type density. The independent population prior supplies the degree-independent diagonal, observed-link selection supplies the degree term, and complementarity supplies the negative adjacency entries of the precision.

No economic argument requires the one-link block Λ_{im} to be positive semidefinite. It is a contribution to the log link propensity, not the precision of a stand-alone Gaussian density. The relevant object is the full graph-level precision $\mathbf{Q} = \mathbf{Q}_0 + \sum_{(i,m) \in E} \Lambda_{im}$. The independent prior contributes a strictly positive diagonal, and the aggregate matrix is positive definite under the graph-level restriction $|\rho| \lambda_{NB} < 1$. Indeed, coefficient matching gives $\alpha_2 \zeta_2 = \rho^2 < 1$; this is not a contradiction because propriety is a condition on the posterior after combining the link likelihood with the prior.

Accuracy of the sparse-link approximation

We now compare the positive-link GMRF with the selected-type distribution from a complete logistic link pattern. This comparison uses a fixed, finite dyadic risk set D that contains the observed edges E_G of a graph $G = (V, E_G)$. For $d \in D$, write the index in equation (38) as

$$\eta_d(\mathbf{x}) = \kappa + h_d(\mathbf{x}) \quad (53)$$

and collect the logit denominators in

$$U_D(\mathbf{x}) = \sum_{d \in D} \log \left(1 + e^{\kappa + h_d(\mathbf{x})} \right). \quad (54)$$

Let p_G^+ denote the normalized positive-link GMRF in equation (42), and assume links are conditionally independent given types on D . Let p_G^F denote the normalized selected-type density conditional on the complete logistic link pattern, proportional to

$$p_0(\mathbf{x}) \prod_{d \in E_G} \Pr(L_d = 1 \mid \mathbf{x}) \times \prod_{d \in D \setminus E_G} \Pr(L_d = 0 \mid \mathbf{x}).$$

The two densities obey the exact identity

$$p_G^F(\mathbf{x}) = p_G^+(\mathbf{x}) \frac{e^{-U_D(\mathbf{x})}}{\mathbb{E}_{p_G^+}[e^{-U_D}]}. \quad (55)$$

Thus the sparse-link approximation drops one common, type-dependent denominator term rather than approximating each observed link in isolation.

Let θ collect all model parameters other than κ . For the Gaussian outcome model, define the positive-link posterior

$$p_{G,y}^+(\mathbf{x}; \theta) = \frac{p(y \mid \mathbf{x}; \theta) p_G^+(\mathbf{x}; \theta)}{p_G^+(y; \theta)}. \quad (56)$$

Write $\ell_G^-(y; \theta) = \log p_G^-(y; \theta)$ for the two marginal outcome log likelihoods. We use the moment envelopes

$$A_G(\theta) = \sum_{d \in D} \mathbb{E}_{p_G^+(\theta)}[e^{h_d}], \quad (57)$$

$$B_G(y; \theta) = \sum_{d \in D} \mathbb{E}_{p_{G,y}^+(\theta)}[e^{h_d}], \quad (58)$$

$$\delta_G(\kappa, \theta) = e^\kappa A_G(\theta). \quad (59)$$

For each $d \in D$, let $G \oplus d$ add one copy of d to G . If d is already observed, the added copy is a parallel edge; the non-backtracking operator treats the two copies as distinct directed edges. Define the robust one-edge radius

$$\lambda_{NB}^{(1)}(G; D) = \max_{d \in D} \lambda_{NB}(G \oplus d). \quad (60)$$

This is a theoretical margin for the approximation, not a change to the binary graph used by the estimator.

Proposition 1 (sparse-link approximation). Suppose G and D are fixed and finite, $E_G \subseteq D$, $|\rho| < 1$, and

$$|\rho| \lambda_{NB}^{(1)}(G; D) < 1. \quad (61)$$

Suppose also that the independent population distribution and the Gaussian outcome model are proper. Then

$$\text{KL}(p_G^F \| p_G^F) \leq \delta_G(\kappa, \theta), \quad (62)$$

and, whenever $\delta_G(\kappa, \theta) < 1$,

$$\text{KL}(p_G^F \| p_G^+) \leq -\log\{1 - \delta_G(\kappa, \theta)\}. \quad (63)$$

The corresponding marginal outcome log likelihoods satisfy

$$|\ell_G^F(y; \theta) - \ell_G^+(y; \theta)| \leq e^\kappa \{A_G(\theta) + B_G(y; \theta)\}. \quad (64)$$

Hence both Kullback–Leibler divergences and the outcome log-likelihood error are $O(e^\kappa)$ as $\kappa \rightarrow -\infty$ for a fixed graph and risk set.

The approximation also extends to likelihood derivatives. Let Θ_0 be a compact parameter set bounded away from $|\rho| = 1$, zero scale parameters, and singular error covariance. Suppose there is an $\epsilon_{NB} > 0$ such that

$$\sup_{\theta \in \Theta_0} |\rho(\theta)| \lambda_{NB}^{(1)}(G; D) \leq 1 - \epsilon_{NB}. \quad (65)$$

Suppose the reduced-form coefficients in $h_d(\theta)$ have bounded first and second derivatives on Θ_0 . For all sufficiently negative κ , there are finite constants C_1 and C_2 , independent of κ , such that

$$\sup_{\theta \in \Theta_0} \|\nabla_\theta \ell_G^F - \nabla_\theta \ell_G^+\| \leq e^\kappa C_1, \quad (66)$$

$$\sup_{\theta \in \Theta_0} \|\nabla_\theta^2 \ell_G^F - \nabla_\theta^2 \ell_G^+\| \leq e^\kappa C_2. \quad (67)$$

Suppose, in addition, that both maximizers are interior to Θ_0 and that ℓ_G^+ is strongly concave there, with minimum information eigenvalue at least $I_G > 0$. If $e^\kappa C_2 < I_G$, then

$$\left\| \widehat{\theta}_G^F - \widehat{\theta}_G^+ \right\| \leq \frac{e^\kappa C_1}{I_G - e^\kappa C_2}. \quad (68)$$

The two estimators therefore differ by $O(e^\kappa)$ for a fixed identified graph model. For a sequence of graphs with K observations and a uniform version of equation (65), if $I_G \geq cK$, retain the sample-dependent envelopes C_{1K} and C_{2K} . Sufficient conditions for the difference between the maximizers to vanish are

$$\frac{e^\kappa C_{1K}}{K} \rightarrow 0, \quad \frac{e^\kappa C_{2K}}{K} \rightarrow 0. \quad (69)$$

This statement does not impose the stronger rate $O(e^\kappa/K)$, which would require $C_{1K} = O(1)$. It also does not establish equivalence at a sampling-distribution rate without correspondingly stronger rate conditions.

Proof. First, $\log(1+u) \leq u$ gives

$$0 \leq U_D(\mathbf{x}) \leq e^\kappa \sum_{d \in D} e^{h_d(\mathbf{x})}. \quad (70)$$

Let $Z_G = \mathbb{E}_{p_G^+}[e^{-U_D}]$. The density ratio in equation (55) gives

$$\text{KL}(p_G^+ \| p_G^F) = \mathbb{E}_{p_G^+}[U_D] + \log Z_G \leq \delta_G, \quad (71)$$

$$\text{KL}(p_G^F \| p_G^+) = -\mathbb{E}_{p_G^F}[U_D] - \log Z_G \leq -\log(1 - \delta_G), \quad (72)$$

where the second inequality uses $Z_G \geq 1 - \mathbb{E}_{p_G^+}[U_D] \geq 1 - \delta_G$. Integrating equation (55) against the outcome density gives

$$\ell_G^F(y; \theta) - \ell_G^+(y; \theta) = \log \mathbb{E}_{p_{G,y}^+}[e^{-U_D}] - \log \mathbb{E}_{p_G^+}[e^{-U_D}]. \quad (73)$$

Jensen's inequality and equation (70) then imply equation (64).

Second, the Gaussian moments in equation (59) are finite when every once-tilted precision is positive definite. Under the coefficient matching,

$$\mathbf{Q}_G + \Lambda_d = \mathbf{Q}_{G \oplus d}. \quad (74)$$

Equation (61) and the same non-backtracking argument as in equation (21) therefore imply $\mathbf{Q}_G + \Lambda_d \succ 0$ for every $d \in D$. Every expectation of e^{h_d} times a polynomial in $\mathbf{x}$ is finite. Conditioning on the Gaussian outcomes adds the positive-semidefinite precision term in equation (29), so the same conclusion holds under $p_{G,y}^+$.

A forest is an immediate special case rather than a necessary assumption. If G is a forest, adding one dyad either preserves a forest or creates one cycle, including the length-two cycle formed by a parallel copy. Hence $\lambda_{\text{NB}}^{(1)}(G; D) \leq 1$, and $|\rho| < 1$ guarantees equation (61).

Third, let q_θ denote either p_G^+ or $p_{G,y}^+$, and define

$$J_q(\theta) = \log \mathbb{E}_{q_\theta}[e^{-U_D(\theta)}]. \quad (75)$$

Writing $\mathbf{s}_q = \nabla_\theta \log q_\theta$ and $\mathbf{H}_q = \nabla_\theta^2 \log q_\theta$, differentiation under the integral gives

$$\nabla_\theta J_q = \mathbb{E}_{q_\theta^F}[\mathbf{s}_q - \nabla_\theta U_D], \quad (76)$$

$$\nabla_\theta^2 J_q = \mathbb{E}_{q_\theta^F}[\mathbf{H}_q - \nabla_\theta^2 U_D] + \text{Cov}_{q_\theta^F}(\mathbf{s}_q - \nabla_\theta U_D), \quad (77)$$

where $q_\theta^F \propto q_\theta e^{-U_D}$. Moreover,

$$\|\nabla_\theta U_D\| \leq e^\kappa \sum_{d \in D} e^{h_d} \|\nabla_\theta h_d\|, \quad (78)$$

$$\|\nabla_\theta^2 U_D\| \leq e^\kappa \sum_{d \in D} e^{h_d} (\|\nabla_\theta^2 h_d\| + \|\nabla_\theta h_d\|^2). \quad (79)$$

The Gaussian identities $\mathbb{E}_{q_\theta}[\mathbf{s}_q] = 0$ and $\mathbb{E}_{q_\theta}[\mathbf{H}_q] + \text{Cov}_{q_\theta}(\mathbf{s}_q) = 0$ cancel all terms that do not contain the correction. For sufficiently negative κ , the normalizing expectation for the change from q_θ to q_θ^F is uniformly bounded away from zero. The remaining terms are bounded by the once-tilted Gaussian polynomial moments established above. Compactness of Θ_0 makes these bounds uniform and yields equation (67).

Finally, the Hessian perturbation bound leaves ℓ_G^F strongly concave with minimum information eigenvalue at least $I_G - e^\kappa C_2$. Evaluate its score at $\widehat{\theta}_G^+$ and apply the mean-value theorem between the two interior maximizers to obtain equation (68). $\square$

Recovering relative link propensities

For one potential pair, Bayes' rule gives

$$p(a, z \mid L = 1) = \frac{\Pr(L = 1 \mid a, z) p_0(a) p_0(z)}{\Pr(L = 1)}. \quad (80)$$

Rearranging gives

$$\Pr(L = 1 \mid a, z) \propto \frac{p(a, z \mid L = 1)}{p_0(a) p_0(z)}. \quad (81)$$

The omitted constant is the unconditional link probability. Data on linked pairs identify relative link propensities across type pairs; an absolute link probability additionally requires the population link rate or data on nonlinks.

Under the sparse-link approximation, equation (81) has log density ratio $\eta(a, z)$ up to a constant. Coefficient matching maps the covariance parameters into the ratios $1/\tau$, α_2/τ , and ζ_2/τ . Normalizing the complementarity coefficient on az to one then gives $(\tau, \alpha_2, \zeta_2)$ through equation (49). A degree-dependent mean could identify the two additional linear-index coefficients if its outcome loadings have sufficient rank; this is a property of the extended model, not an additional empirical result reported here. The intercept and absolute link rate remain outside the positive-link GMRF likelihood. Proposition 1 bounds the likelihood and derivative errors only under its stated risk-set, intensity, moment, and spectral conditions. A small non-backtracking radius alone does not justify dropping the complete-logit correction or show that empirical relative propensities are unaffected.